The documentation and process conversion measures necessary to comply with this revision shall be completed by

INCH-POUND

MIL-S-19500/297B
10 MAR 1989
SUPERSEDING
MIL-S-19500/297A
6 October 1969

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
TYPES 1N1184, 1N1186, 1N1188, 1N1190, 1N3766, 1N3768,
1N1184R, 1N1186R, 1N1188R, 1N1190R, 1N3766R, 1N3768R,
JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon semiconductor power rectifier diodes. Three levels of product assurance are provided for each device as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (DO-5).

1.3 Maximum ratings.

Type	V_R(non-rep)	V_R	V_{RM}	I_O 1/ T_C = 150°C	i_f(surge) at 1/120 s T_C = 150°C	Barometeric pressure (reduced)	t_{rr}
	V(pk)	V dc	V (pk)	A dc	A	mmHg	µs
1N1184	120	100	100	35	500	8	10
1N1186	240	200	200	35	500	8	10
1N1188	480	400	400	35	500	8	10
1N1190	720	600	600	35	500	16	10
1N3766	960	800	800	35	500	30	10
1N3768	1200	1000	1000	35	500	54	10

1/ Derate linearly 1.4 A dc/°C between T_C = 150°C to T_C = 175°C.

Storage temperature: T_C = -65°C to +175°C

Operating temperature: T_C = -65°C to +150°C

1.4 Primary electrical characteristics. $R_{\theta JC}$ = .8°C/W maximum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein. Lot accumulation period shall be three months in lieu of six weeks.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

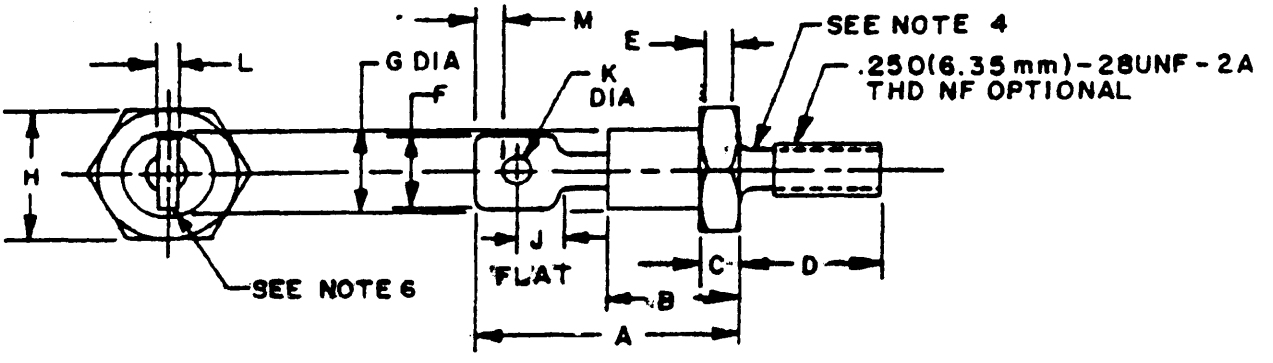
3.4 Marking. Marking shall be in accordance with MIL-S-19500, except at the option of the manufacturer, the following marking may be omitted from the body of the rectifier (diode).

3.4.1 Polarity. The polarity shall be indicated by a graphic symbol with the arrow pointing toward the negative end for forward bias. The reversed units shall also be marked with an R following the last digit in the type number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

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Dimensions				
Dim	Inches		Millimeters	
	Min	Max	Min	Max
A	---	1.000	---	25.40
B	---	.450	---	11.43
C	.115	.200	2.93	5.08
D	.422	.453	10.72	11.50
E	.060	---	1.53	---
F	.250	.375	6.35	9.52
G	---	.667	---	16.94
H	.667	.687	16.95	17.44
J	.156	---	3.97	---
K	.140	.175	3.56	4.44
L	---	.080	---	2.03
M	.030	---	.77	---

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Units must not be damaged by torque of 30 inch-pounds applied to .250-28 UNF-2B nut assembled on thread.
4. Diameter of unthreaded portion .249 (6.32 mm) max and .220 (5.59 mm) min.
5. Complete threads to extend to within 2-1/2 threads of seating plane.
6. Angular orientation of this terminal is undefined.
7. Max pitch diameter of plated threads shall be basic pitch diameter .2268 (5.76 mm) reference FED-STD-H28 (Screw Thread Standards For Federal Services)
8. A chamfer or undercut on one or both ends of the hex portion is optional; minimum base diameter at seating plane .600 inch (15.24 mm).

FIGURE 1. Physical dimensions.

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein. Tests in either polarity shall be sufficient to obtain qualification approval of both polarities.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels
1/	Surge (see 4.3.1) Thermal response (see 4.3.2)
4	Not applicable
9	V_{F2} and I_{R1} 2/
10	MIL-STD-750, method 1038, test condition A; t = 48 hrs
11	Subgroup 2 of table I herein, V_{F2} and I_{R1}; ΔV_{F2} = 1 V(pk) from initial value; ΔI_{R1} = ±40 µA dc or 100 percent from the initial value, whichever is greater. 3/
12	Not applicable
13	Not applicable

1/ Surge shall precede thermal response. These tests shall be performed anytime after screen 3 and before screen 9.
2/ I_{R1} measurement shall not be indicative of an open condition.
3/ PDA of screen 13 shall apply to screen 11.

4.3.1 Surge current. Surge current, see MIL-STD-750, method 4066. I_O = 0; $V_{RM(W)}$ = 0; I_{FSM} = 500a; six surges; T_A = 25°C, t_p = 8.3 ms, one surge per minute maximum.

4.3.2 Thermal response (ΔV_F measurements). The ΔV_F measurements shall be performed in accordance with method 3101 of MIL-STD-750. The ΔV_F conditions and maximum V_F limit shall be derived by each vendor. The chosen ΔV_F measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen ΔV_F values shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. Heating current (I_H) ≥ rated I_O. t_p = 250 ms.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table V herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500, and table II herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 DC intermittent operation life. A cycle shall consist of an "on" period, when power is applied suddenly, not gradually, to the device for the time necessary to achieve a delta case temperature of 85°C +15°C -5°C followed by an "off" period, when the power is suddenly removed, for cooling. Auxiliary (forced) cooling is permitted during the off period only. $30\ s \leq t_{Heating} \leq 60\ s$, $P = V_F \times I_F$ or $P = V_{FPK} \times I_{AV}$ if using sine wave current. DC full wave current (or equivalent half sine wave current) shall be used for the power required during the on period. The test power, or current, shall be at least the free air rating. Within the time interval of 50 cycles before to 500 cycles after the termination of the test, the sample units shall be removed from the specified test conditions and allowed to reach room ambient conditions. Specified end-point measurements for qualification and quality conformance inspection shall be completed within 96 hours after removal of sample units from the specified test conditions. Additional readings may be taken at the discretion of the manufacturer.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual doucment referenced (see 2.1.1).

c. Lead finish may be specified (see 3.3.1).

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions	JAN, JANTX, JANTXV		Min	Max	
Subgroup 1							
Visual and mechanical inspection	2071						
Subgroup 2							
Forward voltage	4011	$I_F = I_{FSM}$ $t_p = 800$ µs		V_{F1}	---	2.5	V dc
Forward voltage	4011	$t_p < 8.3$ ms, duty cycle ≤ 2% pulse $I_f = 110$ a(pk)		V_{F2}	---	1.4	V dc
Reverse current at peak reverse voltage	4016	AC method V_r = Rated V_r (see 1.3)		I_r	---	3	mA(pk)
Reverse current	4016	DC method V_R = Rated V_R (see 1.3)		I_{R1}	---	10	µA dc
Subgroup 3							
High temperature operation		$T_C = 150°C$					
Reverse current	4016	DC method V_R = Rated $V_{R(pk)}$ (see 1.3)		I_{R2}	---	1	mA dc
Subgroups 4, 5, 6, and 7							
Not applicable							

1/ For LTPD numbers, see MIL-S-19500.

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TABLE II. Group B inspection for JANTX and JANTXV.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1 2/				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 2				
Thermal shock (temperature cycling)	1051	Test condition C T(high) = +175°C 10 cycles		
Surge current	4066	T_C = 150°C V_R = Rated V_R (see 1.3) six 1/120s. surges. t_p = 8.3 ms. 1 surge per minute maximum. I_F(surge) = 500 A dc superimposed on I_O = 35 A dc		
Hermetic seal	1071			
Fine leak				
Gross leak				
Electrical measurements		See table V, steps 1, 2, and 3		
Subgroup 3 2/				
DC intermittent operation life	1037	.25 Rated $I_O \leq I_O$ applied $\leq$ Rated I_O (see 4.5.1) 2000 cycles		
Electrical measurements		See table V, steps 1, 2, and 3		
Steady-state DC blocking life	1038	Condition A; 340 hours		
Electrical measurements		See table V, steps 1 and 2		
Subgroup 4				
Decap internal visual (design verification)	2075			

See footnotes at end of table.

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TABLE II. Group B inspection for JANTX and JANTXV - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 5				
Not applicable				
Subgroup 6				
High temperature (nonoperating)	1032	T_A = +175°C minimum		
Electrical measurements		See table V, steps 1 and 2		

1/ For LTPD numbers, see MIL-S-19500.
2/ A separate sample may be pulled for each test.

TABLE III. Group C inspection.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1		
Subgroup 2				
Thermal shock (glass strain)	1056			
Terminal strength	2036			
Bending stress		Test condition B method B; t = 15 s weight = 15 lb		
Terminal torque		Test condition D1 Torque = 3 lb-in; t = 15 s		
Stud torque		Test condition D2 Torque = 30 lb-in t = 15 s		

See footnotes at end of table.

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TABLE III. Group C inspection - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 2 - Continued				
Hermetic seal	1071			
Fine leak Gross leak				
Moisture resistance	1021			
Electrical measurements		See table V, steps 1, 2, and 3		
Subgroup 3				
Shock	2016	500 G		
Vibration, variable frequency	2056	10,000 G		
Constant acceleration	2006			
Electrical measurements		See table IV, steps 1 and 2		
Subgroup 4				
Salt atmosphere (corrosion)	1041			
Subgroup 5				
N/A				
Subgroup 6			5	12/0
DC intermittent operation life	1037	.25 Rated $I_O \leq I_O$ applied $\leq$ Rated I_O (see 4.5.1) 6000 cycles		
Electrical measurements		See table V, steps 1, 3, and 4		
Subgroup 7				
Not applicable				

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IV. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD	Small lot quality conformance inspection n/c
	Method	Conditions		
Subgroup 1			10	6/0
Thermal shock (glass strain)	1056	0°C to 100°C, 100 cycles		
Hermetic seal	1071			
Electrical measurements		See table V, steps 1, 2, and 3		
Subgroup 2			5	12/0
Steady-state DC blocking life	1038 or 1049	Condition A; 1,000 hours		
Electrical measurements		See table V, steps 1 and 4		
Subgroup 3			5/0	1/0
DPA		Photos of cross sections shall be submitted in the qualification report. Vendors shall retain duplicate photos.		
Subgroup 4			10	6/0
Thermal resistance	3101	$R_{\theta JC}$ = .8° C/W maximum I_M = 50 mA I_H = 50 A t_H = 400 ms t_{MD} = 80 µs		
Subgroup 5			10	6/0
Barometric pressure (reduced)	1001	15mm Hg = Pressure t = 60 s. While the test is being performed I_R shall be monitored and shall not exceed 35 mA dc. T_C = 25°C		

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TABLE V. Groups A, B, and C electrical end-point measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	i_F = 110 a(pk) (pulse); pulse width = 8.3 ms, duty cycle = 2% maximum	V_{F2}	---	1.40	V(pk)
2.	Reverse current	4016	DC method V_R = Rated V_R(dc) (see 1.3)	I_{R1}	---	10	µA dc
3.	Thermal response	3101	See 4.3.2	ΔV_F			mV dc
4.	Reverse current	4016	DC method V_R = Rated V_R(dc) (see 1.3)	I_{R3}		20	µA dc

1/ Devices which exceed the group A limits for this test shall be rejected.

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6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's part numbers. This information in no way implies that manufacturer's part numbers are suitable as a substitute for the military part or identifying number.

Military part or identifying number	Manufacturer's CAGE code	Manufacturer's and user's part number

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 85
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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